



**CS411W Team Bronze
Spring 2026
Prototype 1**

Spoiler Alert
‘No Surprises at the Table’

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Meet the Team!



Parker Davis
Project Lead/Database



Shan Ibanez
Backend



Joshua Adkins
tbd



Parker Mullenax
Frontend Developer



Jacqueline Simmons
Frontend Developer



Mark Stoegbauer
Frontend
Developer/Webmaster

The Problem

Food recalls happen all the time, however, there isn't an adequate reporting system overall. It is vital that everybody receives topical recall reports promptly, yet many do not. This is a major safety concern for consumers, as companies don't take proper precautions.

Despite official resources being a thing, their own notifications are restricted to email, not an optimal form in the age of the smartphone. Existing apps also fall short, not being personalized or appealing enough to the average user to be noteworthy.



Solution Statement

Spoiler Alert: An innovative web & mobile application that delivers personalized, immediate, and trustworthy food recall alerts.



- ★ 🕒 **Speed & Urgency** → Instant notifications (push, email).
- ★ 📝 **Clarity** → Simple, consumer-friendly language for critical information.
- ★ 🎯 **Personalization** → Allergy filters, barcode/receipt scanning, custom channels.
- ★ 🔒 **Trust** → Direct FDA/USDA data in one reliable platform.

Result: Families, caregivers, and vulnerable individuals stay safe — no surprises at the table.

Changes to Scope

- Expanded NodeJS library to include Express and EJS.
- Likely to downgrade Barcode Lookup's importance for the demo due to price worries. Will attempt to use free demo, but unsure how far that will take us.
- Database is undergoing restructuring to allow data to actually be stored in an usable manner.

Category	Feature	Description	Progress
Account Management	Account Registration with Email	Users should be able to create new accounts on the signup page. This will require an email and a secure password.	Not Started
	Account Login	Users should be able to access their created account through the login page. This will require their email and the password they used to signup.	Not Started
	Employee Account Creation	Certain employees should be able to create employee accounts, which will allow employees to correct product and recall information.	Not Started
Notifications	Email Alerts	Users should receive emails detailing any recalls that their settings allow them to receive. By default, this will email the user about all of them.	Not Started
	Alert Filtering	Users should be able to have filters set up to block the receiving of alerts that they do not want.	Not Started

Feature Table 1



Category	Feature	Reasoning	Progress
Item Tracking	Inventory Viewing	Users should be able to view products that they have “scanned” (registered) with the application.	In Progress
	Inventory Item Adding	Users should be able to add an inventory to their inventory by entering a UPC into a form.	Not Started
	Inventory Item Removal	Users should be able to remove a product from their inventory if they do not want to be alerted for recalls about that product anymore.	Not Started
Searching	Inventory Product Search	Users should be able to search their inventory by product name or brand.	Not Started
	Keyword Recall Search	Users should be able to search the homepage recall list by recall reason or product.	Not Started

Feature Table 2

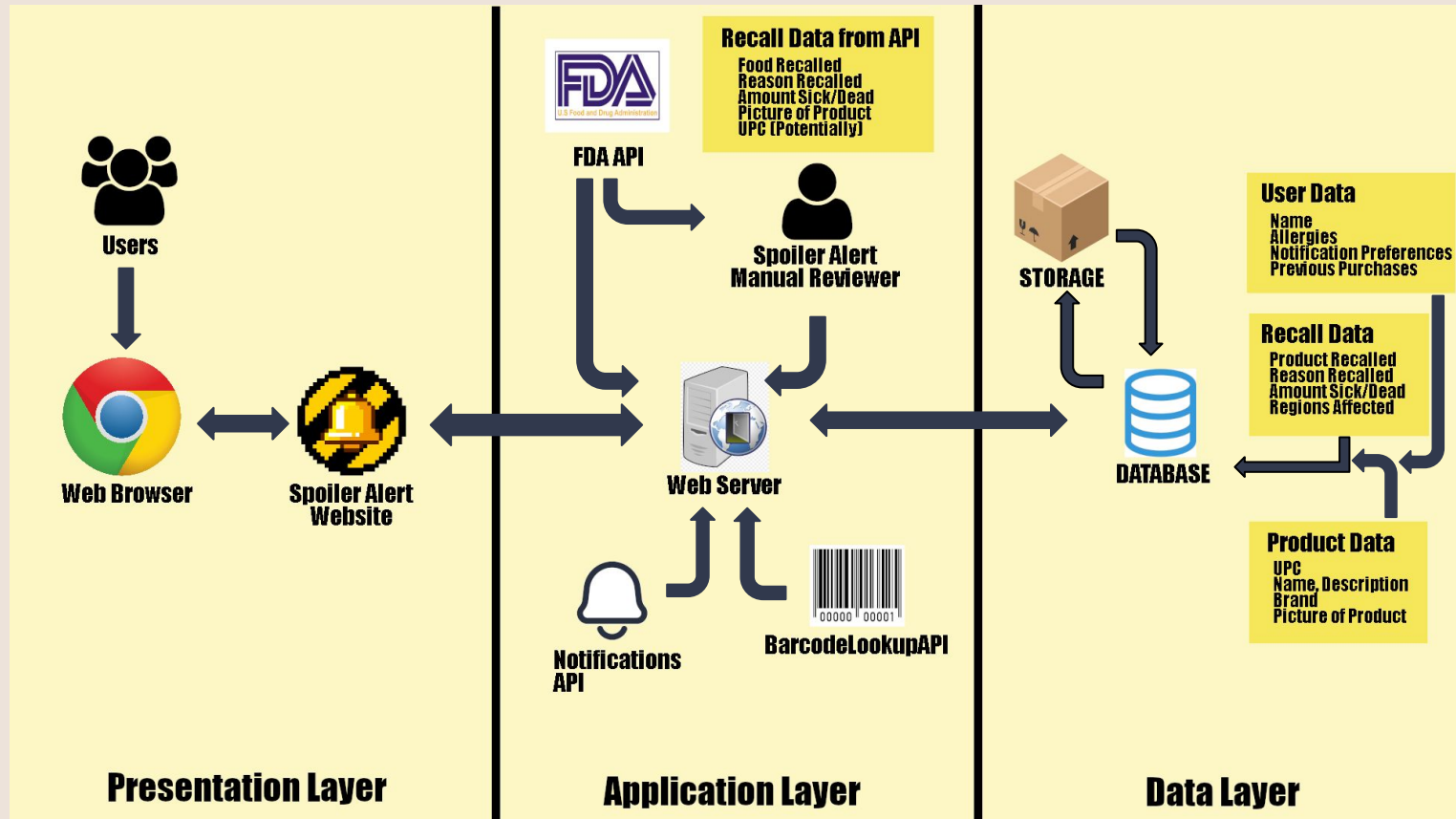


Category	Feature	Reasoning	Progress
Homepage	Recall Timeline	Users should get a list of recent recalls on their homepage.	In Progress
	Filter Recalls	Users should be able to filter the recalls on their homepage in regards to their filtering settings. This can take the form of filtering out mixed in allergens that do not matter to you or filtering out product types entirely.	Not Started
	Recall Details	Users should be able to click on a recall on their homepage to view more information about the recall, including a link to official sources.	Not Started
Recall Reviewer Dashboard	Recall UPC Linking	Employees should be notified of a product that needs a UPC but does not have one yet.	Not Started
	Recall UPC Linking	Employees should be able to add a UPC to a product that does not have one yet.	Not Started
	Manual Recall Editing	Employees should be able to adjust recall information in the event our algorithms do not get all the information they need.	Not Started

Feature Table 3



Major Functional Components

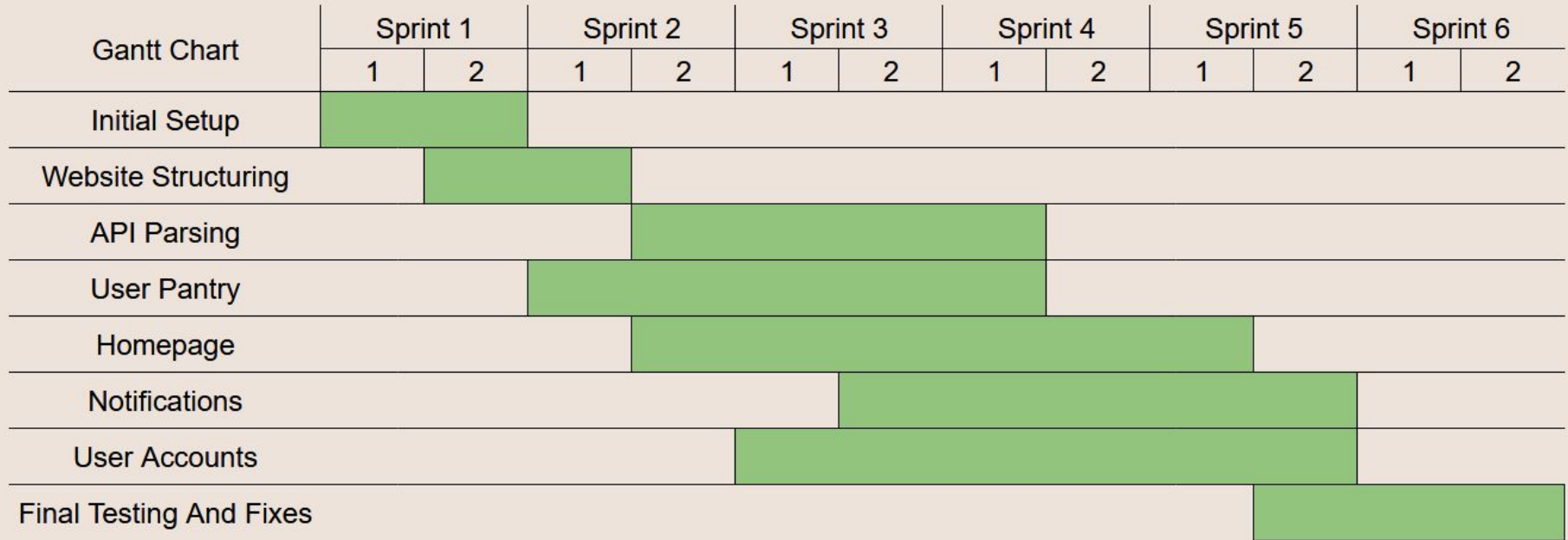


Development Tools



Communication	Discord
Version Control	Git / GitHub
IDE	VSCode
Collaboration	ClickUp
Front End	Bootstrap / EJS
Back End	NodeJS
Database	MySQL

Sprint Breakdown



Sprint Breakdown (2 Week Increments)

Sprint 1

- ❑ Setup code repositories
- ❑ Prepare development tools and frameworks
- ❑ Setup initial database, referencing initial schema
- ❑ Prepare initial pages for the website

Sprint 2

- ❑ Begin parsing openFDA JSON data.
- ❑ Automatically load parsed recall data into database.
- ❑ Load and display recall data into homepage feed.
- ❑ Load and display product data into pantry.
- ❑ Load different product data based on user id.

Sprint 3

- ❑ Filter homepage feed with chosen filters.
- ❑ Filter pantry with chosen filters.
- ❑ Begin work on email notification system.
- ❑ Begin work on user accounts. At least saving new user data.

Sprint Breakdown (2 Week Increments)

Sprint 4

- ❑ Notification customization based on “User Recall Filter” table
- ❑ Ensure usage of functionality of email notifications
- ❑ Finish formatting and display of database entries for users on Pantry page

Sprint 5

- ❑ Finish user accounts
- ❑ Complete email notification pipeline
- ❑ Finish formatting recall listings with detailed information
- ❑ New recall handling

Sprint 6

- ❑ Final UI polish and cleanup
- ❑ Optimize performance
- ❑ Final testing and documentation

User Risks



Risk Factors:

R1: Users may suffer from alert fatigue, from recalls concerning products not relevant to the user.

R2: Users may forget to turn push notifications on, causing a delay in information.

R3: Regional differences may cause misinformation.

R4: Users may not want to share their personal medical information with the app.

Mitigations:

M1: Spoiler Alert customizes reports, so the user only gets notified of the reports that are relevant to them.

M2: Spoiler Alert will alert users that their notifications are off whenever they open the app.

M3: Spoiler Alert will ensure that the regions affected will be one of the first things users see when looking at a recall.

M4: Spoiler Alert will ensure users that the app will encrypt and responsibly store the data. Users may also continue to use the app without entering this data at all.

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe					
	High			R1	R4	
	Medium		R3	R2		
	Low		M1 M2	M4		
	Very Low	M3				

Technical Risks



Risk Factors:

R1: Certain stores may have unique product code formats on their receipts that the app will have trouble looking up.

R2: Low user camera quality can result in unreadable scanning

R3: APIs may have downtime or contain broken data, slowing down or stopping Spoiler Alert's ability to report.

R4: Most recall APIs (FDA, USDA, CPSC) do not include UPCs in their metadata, making it difficult for Spoiler Alert to match scanned product barcodes to active recalls.

Mitigations:

M1: Spoiler Alert will work with the bigger stores to make those product codes searchable for recall logging.

M2: In the case camera quality makes the receipt unreadable, the user can manually input the product code.

M3: If APIs for any reason are unreadable, the servers will attempt to read from the API a few more times. If this still fails, sources will be manually reviewed by staff until the API works once more.

M4: Spoiler Alert will maintain its own UPC database from manufacturer sites, official recall web pages, and product catalogs when the recall APIs don't include it.

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe			R3		
	High			R1		
	Medium	M3				
	Low		R2 M1			
	Very Low	M2				

Legal/Security Risks



Risk Factors:

- R1:** Personal information, such as medical and payment information, can be subject to data leaks.
- R2:** Companies may try to litigate Spoiler Alert for reports that damage their image.
- R3:** The app may collect sensitive personal and medical information without consent.
- R4:** User purchases a product that gets recalled later on. SpoilerAlert may not connect the purchase to the recall and will fail to inform the user.

Mitigations:

- M1:** SpoilerAlert uses up-to-date security protocol to ensure against data attacks.
- M2:** Companies should be held accountable for harm regardless of litigation.
- M3:** App informs and gets users consent with terms of service and privacy policy agreement upon account creation.
- M4:** Popup disclaimer and user agreement to warn about possible missed recalls and limit SpoilerAlert's liability.

Impact

		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Severe				R4	
	High	R1			R3	
	Medium			M4		
	Low	M1				R2
	Very Low		M2 M3			

Next Steps: Parker Davis

Account Management - I will be working with the frontend team to dynamically show inventory data to a user specific page.

Automatic Emails - I will start work on sending out automated emails with new recall information.

Team Management - I will be improving our project management with more frequent updates to our Kanban boards and more personal updates.

Next Steps: Mark Stoegbauer

Improve and Update Team Website - Reformat some of the pages in the team website and add any new information and deliverables.

Research and Draft Page Layouts - Help with work on the visuals of the website, specifically the homepage and login.

Testing and Data Server Hosting - I will be helping test the SQL database as well as setting up a remote database for the team on a personal server.

Next Steps: Jackie Simmons

Update recall page to improve user interface - Change formatting for more user friendly experience.

Format homepage to display user specific data - Add features for user to add their personal information.

Update pantry pages to improve user experience - Change design for user to add personal preferences for personalized notifications

Next Steps: Parker Mullenax

Improve Recall Card Display and Layout - Improve formatting and add links to recall pages.

Improve Pantry Display and Layout - Improve formatting when displaying pantry items.

Implement Filtering UI - Add recall filtering controls.

Support User Account Interface - Create the registration form and login form and display user specific content.

Begin Email Preference UI - Begin design for notification preferences page.

Next Steps: Shan Ibanez



Develop Homepage and Pantry Endpoints - Build the API routes needed to query and supply the recall data for the homepage feed and product data for the pantry.

Automate Database Loading - Create the logic to automatically insert the parsed recall data into our database.

Next Steps: Joshua Adkins

Implement API - Service plans to use openFDA API for weekly published recall reports.

Investigate DB Schema - Service needs an extensible data store for API response storage and retrieval for users.

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Any Questions?

